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Studierichting: Informatica
Oefeningenreeks 4A nr. 62.3

$$
\begin{aligned}
I &= \int \frac{\sqrt{1-x^2}}{x} dx \\
&\overset{x=\cos y}{=} \int \frac{\sqrt{1-\cos^2 y}}{\cos y}(-\sin y) dy \\
&= -\int \frac{\sin^2 y}{\cos y} dy = -\int \frac{1-\cos^2 y}{\cos y} dy = -\int \frac{1}{\cos y} dy + \int \cos y dy \\
&= -\ln\left|\tan\left(\frac{y}{2}+\frac{\pi}{4}\right)\right| + \sin y + C \\
&= -\ln\left|\tan\left(\frac{y}{2}+\frac{\pi}{4}\right)\right| + \sqrt{1-\cos^2 y} + C \\
&\overset{y=\operatorname{Bgcos} x}{=} -\ln\left|\tan\left(\frac{\operatorname{Bgcos} x}{2}+\frac{\pi}{4}\right)\right| + \sqrt{1-x^2} + C
\end{aligned}
$$

References